

Section 24: HSRP

Layer: L3 | Reference: Odom Vol2 Ch16: First Hop Redundancy Protocols | *Original study notes — CCNA 200-301*

HSRP (Hot Standby Router Protocol) provides gateway redundancy by letting two or more routers share a virtual IP address. If the active router fails, a standby router takes over transparently, with no reconfiguration needed on end hosts.

Key Concepts

- HSRP groups present a single virtual IP (and virtual MAC) as the default gateway to hosts on a LAN
- One router is Active (forwards traffic for the virtual IP); one is Standby (ready to take over); others (if any) are Listen state
- Default Hello timer = 3 seconds; default Hold timer = 10 seconds. If no Hello is heard within the Hold time, Standby becomes Active
- Election is based on priority: highest priority wins (default 100). Tie is broken by highest configured IP address
- 'preempt' allows a router to reclaim the Active role once it comes back online with a higher priority than the current Active - without preempt, the original router stays Standby even after recovering
- Interface tracking lets HSRP proactively decrement a router's priority if a tracked upstream interface goes down, triggering failover before the LAN-facing interface itself fails
- HSRPv1 supports group numbers 0-255 and IPv4 only; HSRPv2 supports group numbers 0-4095 and adds IPv6 support
- VRRP (Virtual Router Redundancy Protocol) is the IEEE open-standard equivalent to HSRP
- GLBP (Gateway Load Balancing Protocol) is a Cisco-proprietary alternative that allows active load balancing across multiple gateways instead of one router sitting idle in standby

Command Reference

Command	Purpose
<code>standby <group> ip <virtual-ip></code>	Configure the virtual IP for an HSRP group on an interface
<code>standby <group> priority <0-255></code>	Set this router's priority for the group (default 100; higher wins)
<code>standby <group> preempt</code>	Allow this router to reclaim Active status once it has the highest priority again
<code>standby <group> track <intf> [decrement <value>]</code>	Reduce priority if a tracked interface goes down, triggering failover
<code>standby version 2</code>	Switch the group to HSRPv2 (wider group ID range, IPv6 support)
<code>show standby brief</code>	Quick view of HSRP group state (Active/Standby), priority, and virtual IP

Configuration Walkthrough

Configure HSRP with preempt and interface tracking on the primary router

```
R1(config)#interface GigabitEthernet0/0
R1(config-if)#standby 1 ip 192.168.1.1
R1(config-if)#standby 1 priority 150
R1(config-if)#standby 1 preempt
R1(config-if)#standby 1 track GigabitEthernet0/1 decrement 60
```

Matching standby router (lower priority, no need for preempt)

```
R2(config)#interface GigabitEthernet0/0
R2(config-if)#standby 1 ip 192.168.1.1
R2(config-if)#standby 1 priority 100
```

Worked Scenario

Q: R1 was Active, failed, and R2 took over as expected. R1 comes back online with a higher priority than R2, but traffic keeps flowing through R2. Why?

Preempt was not configured on R1. Without 'standby <group> preempt', a router that regains a higher priority will NOT automatically reclaim the Active role; it stays in Standby even though it would otherwise 'deserve' to be Active. Add 'standby 1 preempt' on R1 to allow it to take back the Active role once its priority is again the highest in the group.

Common Mistakes / Gotchas

- Forgetting 'preempt' and assuming HSRP will automatically restore the 'preferred' router to Active after it recovers
- Setting priorities equal on both routers without realizing the tiebreaker (highest configured IP) may not match design intent
- Confusing HSRP virtual IP (gateway address for hosts) with the routers' own physical interface IPs

Exam Tips

- ★ Know default Hello=3s / Hold=10s timers cold - commonly tested directly
- ★ Preempt + track is a classic exam config pattern for proactive failover
- ★ Remember GLBP is the Cisco answer to 'can we load-balance across both gateways instead of one sitting idle'